

ASL-DVS Node Control Dashboard - Summary

Overview

This is a comprehensive Python-based web dashboard (v8.0.3) for controlling ASL (AllStar Link) and DVS (Digital Voice Switch) radio systems. It provides a browser-based interface to manage radio connections, talkgroups, reflectors, and system configuration.

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****Key Update (v8.0.3):**** Startup no longer forces ASL disconnect or DVS retune on every process restart—only on genuine fresh system boots.

Core Features

System Support

- ****ASL Node Management**** - Control AllStar Link connections and nodes
- ****Digital Modes**** - DMR, STFU, YSF, P25, NXDN, D-STAR, FCS, XLX, M17
- ****EchoLink Support**** - Connect to EchoLink nodes via ASL gateway
- ****Multiple Bridge Nodes**** - Support for Digital Voice Bridge and M17 Bridge slots

Dashboard Capabilities

****Connection Management:****

- Switch between ASL, Echo, and multiple digital mode tabs
- Quick-tune interface for talkgroups and reflectors
- Per-mode Last Heard links to network dashboards
- Permanent link detection and management
- Radio keyed state monitoring with visual indicators

****Configuration:****

- Edit ASL/Echo node lists
- Manage talkgroup favorites for each mode
- Configure DMR networks with per-network talkgroups
- Save D-STAR, XLX, and M17 reflector lists
- Set custom network dashboard URLs
- Enable/disable tabs per system

****System Features:****

- Real-time uptime display
- System reboot/shutdown controls
- Keyed indicator with color-coded mode display
- Confirmation animations on successful tuning

- Toast notifications for user feedback
- Modal dialogs for critical actions
- SysMon integration probe

API Endpoints

Status & Configuration

- `GET /api/status` - System state (with ETag caching for efficiency)
- `GET /api/config` - Configuration details
- `GET /api/keyed` - Current keyed state and linked nodes
- `GET /api/abinfo` - Analog Bridge info (mode, last tune, TG name)
- `GET /api/sysmon-status` - Check if SysMon bridge is running

Data Endpoints

- `GET /api/talkgroups` - All talkgroups by mode
- `GET /api/asl-nodes` - Configured ASL nodes
- `GET /api/echo-nodes` - Configured Echo nodes
- `GET /api/xlx-reflectors` - XLX reflector list
- `GET /api/m17-reflectors` - M17 reflector list
- `GET /api/dmr-servers` - Configured DMR networks
- `GET /api/dmr-server-tgs` - TGs per DMR network
- `GET /api/dmr-ready` - DMR connection ready status

Action Endpoints

- `POST /api/action` - All control actions (tune, connect, switch tabs, save config, etc.)
- `GET /api/event` - External event trigger (for shell scripts)

Key Functions

Configuration Management

- `load\_conf()` - Load/parse config file with backup recovery
- `action\_save\_config()` - Update system settings
- `action\_save\_all()` - Save nodes, talkgroups, and echo nodes
- `action\_save\_dmr\_server\_tgs()` - Save DMR server configurations
- `action\_save\_xlx()` / `action\_save\_m17()` - Save reflector lists

Connection Control

- `action\_switch\_tab()` - Switch radio mode/page
- `action\_tune()` - Connect to talkgroup/reflector
- `action\_quick\_tune()` - Fast tune with minimal parameters

- `action\_retune\_tab()` - Retune current page
- `action\_disconnect\_all()` - Emergency disconnect + service restart

ASL Operations

- `action\_asl\_connect()` - Connect to ASL node
- `action\_echo\_connect()` - Connect to EchoLink node
- `action\_disc\_perm\_link()` - Clear external permanent links
- `\_poll\_asl\_state()` - Monitor link status and multi-link cleanup

Digital Mode Operations

- `action\_dstar\_connect()` / `action\_xlx\_connect()` - D-STAR/XLX reflector connect
- `action\_m17\_connect()` / `action\_m17\_disconnect()` - M17 reflector operations
- `action\_switch\_dmr\_server()` - Select DMR network and wait for ready
- `\_enter\_digital()` - Setup digital mode with bridge linking

System Detection

- `\_detect\_asl()` - Check for Asterisk/rpt.conf
- `\_detect\_dvs()` - Find dvswitch.sh
- `\_detect\_stfu()` - Check STFU service/binary
- `\_detect\_echo()` - Probe for EchoLink module
- `\_detect\_m17()` - Check USRP2M17 service
- `\_is\_fresh\_boot()` - Detect boot vs. process restart (via `/proc/sys/kernel/random/boot\_id`)

Monitoring & Status

- `\_link\_poll\_loop()` - Continuous ASL state polling (1Hz)
- `\_bridge\_watchdog()` - Auto-reconnect bridge nodes if down
- `action\_get\_keyed()` - Return keyed status and perm link info
- `build\_status()` - Aggregate current system state

Configuration I/O

- `\_parse\_conf()` - Parse ini-style config with sections
- `\_write\_conf()` - Atomic write with backup rotation
- `\_atomic\_write()` - Fsync-safe file writes
- `\_rotate\_backup()` - Keep 3 most recent backups

Startup & Initialization

- `startup()` - Cold-start setup (boot detection, connection preservation)
- `\_is\_fresh\_boot()` - Differentiate boot from restart
- `\_dvs\_startup\_init()` - Initialize DVS in DMR mode
- `action\_handle\_event()` - Handle external event triggers

Installation

System Requirements

- **OS:** Linux (Raspberry Pi OS, Debian, Ubuntu, etc.)
- **User:** Root privileges required
- **Python:** 3.6+
- **Dependencies:** systemd (for service management)

Prerequisites

Ensure your system has:

- Asterisk with ASL (AllStar Link) installed
- DVSwitch / MMDVM Bridge (for digital modes)
- Optional: STFU, EchoLink module, M17 bridge

Installation Steps

1. **Automatic Installation (Recommended)**

```
```bash
sudo python3 asl_dvs_dashboard.py --install
```
```

This will:

- Copy the script to `/usr/local/lib/asl_dvs/`
- Create symlink at `/usr/local/bin/asl_dvs_dashboard.py`
- Install systemd service at `/etc/systemd/system/asl_dvs_dashboard.service`
- Enable and start the service
- Print access URLs to the console

2. **Manual Installation**

```
```bash
Copy script
sudo cp asl_dvs_dashboard.py /usr/local/lib/asl_dvs/
sudo chmod 755 /usr/local/lib/asl_dvs/asl_dvs_dashboard.py

Create symlink
sudo ln -s /usr/local/lib/asl_dvs/asl_dvs_dashboard.py /usr/local/bin/asl_dvs_dashboard.py

Install service file
sudo cp asl_dvs_dashboard.service /etc/systemd/system/

Enable and start
sudo systemctl daemon-reload
sudo systemctl enable asl_dvs_dashboard.service
sudo systemctl start asl_dvs_dashboard.service
```
```

3. **Access the Dashboard**

After installation, the service will output access URLs. Default:

- **Local:** `http://localhost:8989`
- **Network:** `http://your-pi-ip:8989`

Configuration

Initial Config

- On first run, creates default config at `/etc/asl\_dvs/asl\_dvs.conf`
- Contains sections for: `[CONFIG]`, `[ASL]`, `[ECHO]`, `[DMR]`, `[STFU]`, `[YSF]`, `[FCS]`, `[P25]`, `[NXDN]`, `[DSTAR]`, `[XLX]`, `[M17]`, `[DMR\_SERVERS]`

Edit Configuration

Use the **EDIT** tab in the dashboard, or manually edit:

```
```bash
sudo nano /etc/asl_dvs/asl_dvs.conf
```
```

Key settings:

```
```ini
[CONFIG]
asl_node=652702 # Your ASL node number
bridge_nodes=1999,, # Digital bridge node (slot 0), M17 bridge (slot 1)
port=8989 # Web UI port
callsign=KD8PGK # Your callsign
enabled_tabs=ASL,DMR,... # Visible tabs
```
```

Backup Config

Backups are auto-saved to `/etc/asl\_dvs/asl\_dvs.conf.YYYYMMDD\_HHMMSS.bak` (3 most recent kept)

Monitoring & Logs

View live logs:

```
```bash
sudo journalctl -u asl_dvs_dashboard -f
```
```

View recent logs:

```
```bash
sudo journalctl -u asl_dvs_dashboard -n 100
```
```

Uninstallation

```
```bash
sudo python3 asl_dvs_dashboard.py --uninstall
```
```

This will:

- Stop and disable the systemd service
- Remove service file
- Remove symlink and installed script
- Leave `/etc/asl\_dvs/asl\_dvs.conf` untouched for recovery

Troubleshooting

****Service won't start:****

```
```bash
sudo systemctl status asl_dvs_dashboard
sudo journalctl -u asl_dvs_dashboard -n 50
```
```

****Can't connect to web UI:****

- Check if service is running: `systemctl status asl\_dvs\_dashboard`
- Verify port not in use: `sudo netstat -tlnp | grep 8989`
- Check firewall: `sudo ufw status`

****Config not loading:****

- Check file permissions: `sudo ls -la /etc/asl\_dvs/`
- Check backups exist: `sudo ls -la /etc/asl\_dvs/asl\_dvs.conf\*.bak`
- Restore from backup if needed

****ASL/DVS not detected:****

- Verify Asterisk running: `ps aux | grep asterisk`
- Verify DVSwitch installed: `which dvswitch.sh` or `ls /opt/MMDVM\_Bridge/dvswitch.sh`
- Check search paths in code (`\_DVS\_SEARCH\_PATHS`)

Performance Notes

- ****Status polling:**** 3-10 second intervals (adaptive based on error rate)
- ****Keyed polling:**** 500ms (continuous monitoring)
- ****Connection limit:**** 24 concurrent clients
- ****Config backups:**** Keep 3 most recent versions

- **Compression:** HTML/JSON auto-gzipped for clients supporting it
- **Caching:** ETag-based 304 Not Modified for status endpoints

Security Considerations

- **Root required:** Must run as root (Asterisk/systemd requirements)
- **Local network only:** No built-in authentication (assume trusted LAN)
- **CSP headers:** Restricts inline script/style execution
- **Input validation:** Sanitizes control characters, validates numeric fields
- **Rate limiting:** Throttles Asterisk commands to prevent flooding

File Structure After Installation

...

/etc/asl_dvs/

- └─ asl_dvs.conf # Main configuration
- └─ asl_dvs.conf.*.bak # Backups (3 kept)

/usr/local/lib/asl_dvs/

- └─ asl_dvs_dashboard.py # Installed script

/usr/local/bin/

- └─ asl_dvs_dashboard.py # Symlink to installed script

/etc/systemd/system/

- └─ asl_dvs_dashboard.service # Systemd service

/run/asl_dvs/

- └─ boot_marker # Boot ID marker (tmpfs)

/var/log/

- └─ (journalctl) # Service logs

...

Version History Note

v8.0.3 (2026-08-22):

- Fixed: Startup no longer forces disconnect/retune on process restart

- Uses ``/proc/sys/kernel/random/boot_id`` to detect genuine boots
- Preserves live ASL links and DVS connections across service restarts
- Only hard-resets on actual system reboot

This dashboard provides professional-grade radio system management with a modern web interface, comprehensive logging, and robust error handling.